

Detecting Chilean Needle Grass with AI-Powered Segmentation

☰ Question 2	
👤 Respondent	z zeeshan khan
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Detecting Chilean Needle Grass with AI-Powered Segmentation (Australia)

Overview

Chilean Needle Grass (*Nassella neesiana*) is a highly invasive weed that poses a serious threat to pastures, native ecosystems, and agricultural productivity. Early and accurate detection is essential for effective management, but identifying this weed across large areas is challenging due to its visual similarity to other grasses.

To tackle this, we developed an AI-based segmentation system capable of precisely identifying Chilean Needle Grass in aerial imagery. By leveraging high-resolution drone data, polygon-based annotations, and advanced deep learning, we created a scalable solution for monitoring and managing this invasive species.

How We Did It

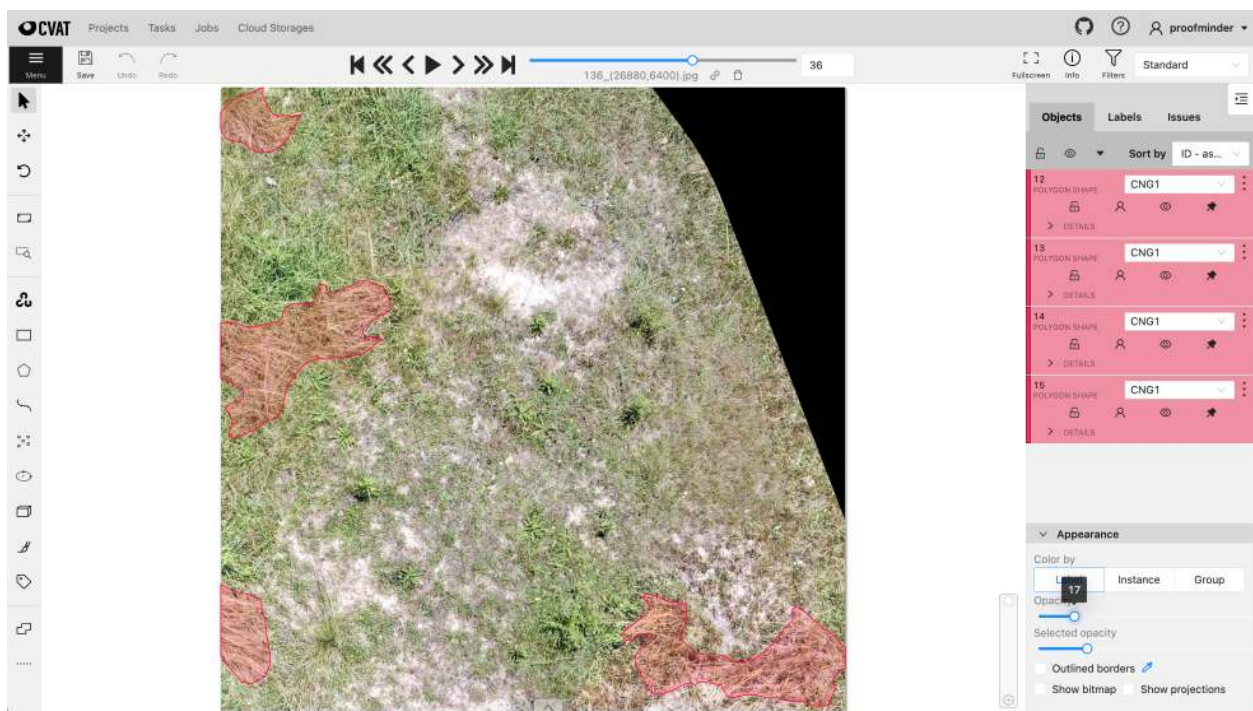
1. Data Collection

We began by capturing high-resolution drone images of the many hectares of target fields at a Ground Sampling Distance (GSD) of 0.5. These individual images were stitched together to form orthomosaics, providing a single, seamless view of large areas. This step was crucial for ensuring spatial continuity and accurate mapping.

To train our segmentation model, we extracted **1280×1280 pixel crops** from the orthomosaics. These crops provided a manageable size for annotation and model input without losing important details.

2. Data Annotation

The cropped images were uploaded to **CVAT (Computer Vision Annotation Tool)**, where we drew detailed **polygons** outlining patches of Chilean Needle Grass. This precise, pixel-level labeling ensured the model learned to distinguish even subtle visual differences from surrounding vegetation.



[Figure 2.1] The annotators are labeling the data

3. Model Development

Using the annotated dataset, we trained a deep learning-based segmentation model. The model's objective was to produce pixel-wise classifications—assigning each pixel in an image to either **Chilean Needle Grass** or **background**.

3. Inference on Orthomosaics

Once trained, we applied the model to the full orthomosaics. The result was a complete, high-resolution segmentation map of Chilean Needle Grass across the

entire area, highlighting infestation zones for further analysis and targeted management.



[Figure 3.1] shows the detections over a large othomosaic file



- **Red Polygons** mark detected Chilean Needle Grasses.

Results & Analysis

- **High-Resolution Segmentation Maps:** Generated outputs clearly show areas affected by Chilean Needle Grass, making it easy to prioritize control efforts.
- **Scalable Analysis:** From small plots to large properties, the workflow scales seamlessly—only requiring updated drone imagery for new surveys.

- **Actionable Data:** Land managers can now quantify the extent of infestation, track changes over time, and make data-driven decisions for eradication strategies.

Benefits of the Approach

- **Precision:** Pixel-level segmentation ensures minimal false positives in visually complex environments.
- **Efficiency:** Eliminates the need for exhaustive ground surveys, saving time and resources.
- **Repeatability:** The pipeline can be reused for ongoing monitoring with updated imagery.

Our segmentation-based approach to Chilean Needle Grass detection demonstrates how AI can transform invasive weed management. By combining drone data, robust annotation, and deep learning, we've built a solution that delivers accurate, large-scale insights—empowering land managers to act faster and more effectively against one of agriculture's persistent challenges.